

Module-1

Lecture-2: Stress & Strain

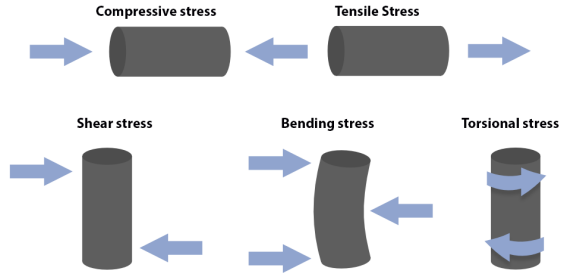
MECHANICS OF SOLIDS

- 1 Stress
- 2 Types of stress
- 3 Strain
- 4 Types of strain
- 5 Elastic constants
- 6 Stress Strain curve

- Internal resistive force/Area
- Condition of equilibrium
- $\sigma = \frac{P}{A}$
- Unit is Pascal $1Pa = \frac{1N}{m^2}$

Types of Stresses

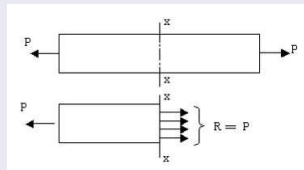
- Normal stress (σ)
 - Tensile stress
 - Compressive stress
- Shear stress (τ)
- Bending stress
- Torsional stress



Normal Stress

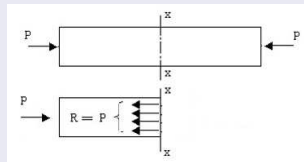
Tensile stress

- As a result of equal and opposite pull
- Increase in dimension along the axis of force
- Decrease in dimension normal to the axis of force



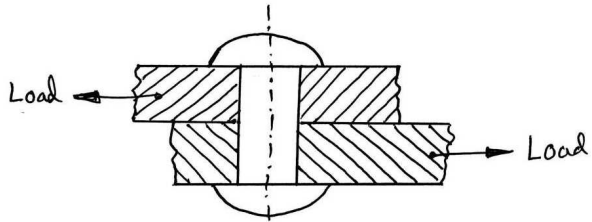
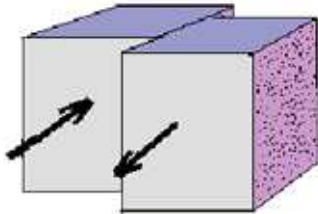
Compressive stress

- As a result of equal and opposite push
- Decrease in dimension along the axis of force
- Increase in dimension normal to the axis of force



Shear Stress

- Two equal and opposite force acting tangential to resisting cross section
- Body will shear off across the section
- Denoted by τ

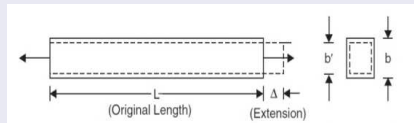


- Result of external force acting on deformable bodies
- Change in dimension/Original dimension
- Dimensionless
- Types
 - Normal strain (ϵ)
 - Shear strain (ϕ)
 - Volumetric strain

Normal strain

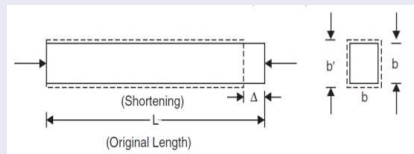
Tensile strain

- As a result of tensile stress
- Increase in length/Original length
- $\epsilon = \frac{\delta L}{L}$



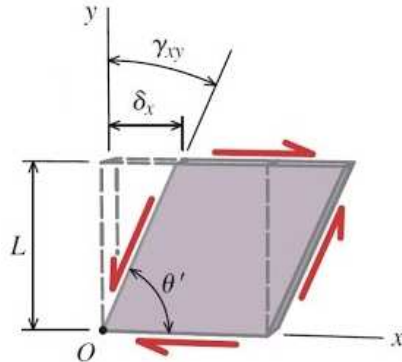
Compressive strain

- As a result of compressive stress
- Decrease in length/Original length
- $\epsilon = \frac{\delta L}{L}$



Shear Strain

- Strain produced by shear stress
- Transverse displacement/height
- Denoted by ϕ
- $\tan\phi = \frac{\delta_x}{L} = \phi$



Volumetric Strain

- Change in volume/Original volume
- Denoted by ϵ_v
- $\epsilon_v = \frac{dV}{v}$

Elasticity

Ability of a body to regain its original shape and size when external force is removed

Elastic limit

For every body, there is a limiting value of force up to which property of Elasticity exist. The value of stress corresponding to this limiting force is called Elastic Limit

Hooke's Law

- Named after 17th-century British physicist Robert Hooke
- When a material is loaded, within its elastic limit, the stress is proportional to the strain

Elastic constants

Modulus of Elasticity (Young's Modulus)

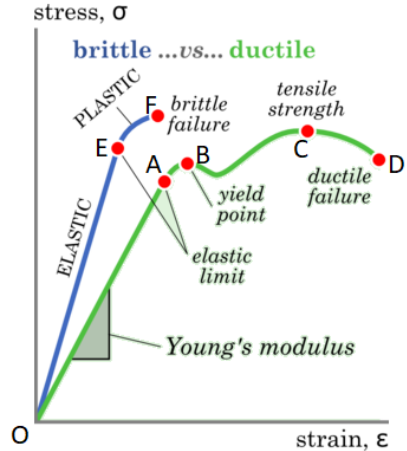
- Ratio of tensile or compressive stress to corresponding strain is a constant within elastic limit. This constant is called Young's modulus
- $E = \frac{\sigma}{\epsilon}$
- E is measured in Pascal
- $E_{Steel} = 200GPa$

Modulus of rigidity (Shear modulus)

- Ratio of shear stress to corresponding shear strain is a constant within elastic limit. This constant is called Shear modulus
- Denoted by C or G or N
- $C = \frac{\tau}{\phi}$
- C is measured in Pascal

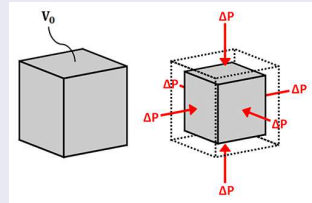
Stress strain curve

- Elastic limit (A)
 - Material shows elastic behaviour
 - Hooke's law
- Yield point (B)
 - Stress remains constant. Strain increases
- Ultimate stress (C)
 - Max stress
 - c/s of material start reducing very fast (necking)
 - Beyond this point stress reduces
- Breaking point (D)
 - Failure occurs



Bulk Modulus

- Normal stress required to produce a unit volumetric strain.
- Denoted by K
- Direct stress/Volumetric strain
- $K = \frac{\sigma}{\epsilon_v}$



Factor of safety

- Ultimate(permissible) stress/Working stress
- $F.S = \frac{\sigma_{ultimate}}{\sigma_{working}}$

*Thank
you!*